Fredrik Pettit

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Education

University of British Columbia — Vancouver, BC
B.Sc. Computer Science, Artificial Intelligence Option — *Expected Graduation: May 2028*

Experience

AI Engineer Intern — May 2026 – Sep 2026
AEM — *Hayward, CA*

- Saved **$16K** a year in staff time with an AI support agent that answers customers in English and German
- Deployed the agent as a FastAPI service on Azure Kubernetes that streams each reply to the customer within 3 seconds and hands anything it can't answer to the help desk through a webhook
- Raised auto-resolution from 32% to **43%** with a RAG pipeline on Azure using hybrid search over 4,000 pages of company docs after tracing most escalations to German and acronym-heavy customer questions
- Caught a 6-point drop in correct answers from a silent model update with an automated 300-case regression eval
- Cut monthly budget reporting from **6+ hours to 5 minutes** with a LangChain agent that turns NetSuite exports into Excel reports through Microsoft Graph API and sends unsure expenses to finance for human review

Software Engineering Intern — Jan 2026 – Apr 2026
ELLUSTROS — *The Blue Mountains, ON*

- Built the FastAPI REST API that recommends best-fit chairs to 300+ users by scoring each chair's ergonomic attributes against their body measurements from the React Native app
- Shipped PostgreSQL schema changes through Alembic without losing user data or breaking older app versions
- Cut failed photo uploads on slow mobile networks by **70%** with presigned URLs that send images straight to S3
- Kept the app and API in sync by testing requests against Pydantic models in GitHub Actions CI/CD

Teaching Assistant | CPSC 310, CPSC 110 — Aug 2025 – Present
UBC Department of Computer Science — *Vancouver, BC*

- Earned a 4.9/5 average student rating while teaching 900+ students in UBC's largest computer science course
- Wrote 250+ randomized JavaScript questions and Python autograders for CPSC 310

Projects

Inventory & Sales Automation Platform | *Java, Spring Boot, Go, React, PostgreSQL, Redis, Pub/Sub, Docker*

- Built a Spring Boot and React app that tracks orders and profit for a personal resale business with **$6K+** in sales
- Automated order entry with a concurrent Go worker that turns eBay, Depop and Grailed sale emails from Gmail and Pub/Sub into orders and skips duplicates by order ID
- Sped up the sales dashboard API **36×** by caching report queries in Redis and clearing them on each new order

YouTube Shorts AI Optimizer | *Python, FastAPI, Claude API, Gemini API, Google Cloud Run, BigQuery*

- Built a Claude and Gemini agent with **100+** users that reviews Shorts and grew a personal channel to **45M** views
- Completed **99.9%** of video reviews at launch with Cloud Tasks retries on idempotent Cloud Run workers
- Cut p95 latency from 95s to 18s by tracing agent steps with OpenTelemetry and parallelizing Claude tool calls

Apex FSD - StormHacks 2026 (Winner) | *Python, PyTorch, OpenCV, PySDL2*

- Won Best Python Hack with a self-driving AI that steers a real wheel in Forza through a 100 Hz PySDL2 loop

GitIntent - UC Berkeley AI LLM Hackathon 2026 | *Python, FastAPI, Redis Stack, TypeScript*

- Helped coding agents solve 8.3% more DeepSWE benchmark tasks with a memory layer of similar past work
- Indexed every code change in Redis with HNSW and BM25 under a SHA-256 hash

Technical Skills

Languages: Java, Python, Go, TypeScript/JavaScript, C++, SQL, C#
Frameworks & Data: Spring Boot, FastAPI, React Native, SQLAlchemy, PostgreSQL, Redis, BigQuery, pytest, Jest
Cloud & DevOps: AWS (S3), GCP, Azure, Kubernetes, Docker, Pub/Sub, GitHub Actions, OpenTelemetry
AI/ML: PyTorch, LangChain, Azure OpenAI, Claude/Gemini APIs, RAG, Azure AI Search